Water Bottle Rocket Project Documentation

Design, Aerodynamic Trajectory Optimization & Practical Launch Analysis

Project Type:	Fluid Dynamics & Aerospace Engineering Demonstration	**Launch Angle:**	~35° (Optimized for Aerodynamic Drag)
Propellant Fill:	25% Water / 75% Compressed Air	**Launch Pad:**	Custom PVC Pipe Frame Structure

1. Executive Summary & Overview

This project covers the design, fabrication, and field evaluation of a high-performance water bottle rocket and its custom PVC launch pad structure. While theoretical ballistic mechanics suggests a 45° launch angle yields maximum distance in a vacuum, real-world aerodynamic drag and pressure thrust profiles require an optimized lower launch angle. In this experiment, an elevation angle of approximately **35°** was selected alongside a **25% water volume fraction** to optimize overall range and stability.

2. Technical Specifications

Subsystem / Parameter	Design Specification	Engineering Rationale
Fuselage Pressure Vessel	PET Bottle Structure with Nose Cone & Stabilizing Fins	High tensile strength to withstand multi-bar pneumatic pressure; low structural weight.
Aerodynamic Nose Cone	Conical / Parabolic Geometry	Reduces frontal pressure drag (C_d) during high-velocity flight phase.
Fin Assembly	3-Fin Symmetrical Airfoil/Flat Design	Shifts the Center of Pressure (CP) behind the Center of Mass (CM) to ensure static stability.
Launch Stand Assembly	Custom PVC Pipe Frame Structure	Lightweight, rigid, cost-effective structure supporting targeted 35° launch elevation and quick-release nozzle locking.
Propellant Ratio	25% Water / 75% Air Volume	Balances reaction mass (thrust) with energy capacity (compressed air volume) for maximum horizontal impulse.
Launch Angle	~35° Horizontal Elevation	Compensates for aerodynamic drag forces to yield maximum horizontal displacement.

3. Mathematical Model & Aerodynamic Analysis

3.1 Thrust Dynamics & Reaction Mass

Water rocket propulsion relies on Newton's Third Law of Motion. Compressed air within the pressure vessel forces water out through the nozzle at high speed:

$$F_{thrust} = \dot{m} \cdot v_e + (P_e - P_0) \cdot A_e$$

Where $\dot{m}$ represents the mass flow rate of water, v_e is the exit velocity governed by Bernoulli's equation ($v_e = \sqrt{2(P - P_0) / \rho}$), and A_e is nozzle cross-sectional area. A **25% water fill** provides optimal impulse duration without overburdening the rocket with dead weight during initial acceleration.

3.2 Trajectory Optimization: 35° vs 45° Angle

In ideal vacuum physics, maximum horizontal range is achieved at $\theta = 45°$ according to the range formula $R = (v^2 \sin 2\theta) / g$. However, in atmospheric flight, air resistance creates a quadratic drag force:

$$F_{drag} = \tfrac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot A$$

Because drag decelerates horizontal velocity continuously, spending more time in higher altitude arcs (higher release angles like 45°) increases total velocity dissipation. By lowering the launch angle to ~**35°**:

- The forward velocity vector ($v_x = v \cdot \cos\theta$) is increased by ~8.2% relative to a 45° angle.
- The total time of flight is reduced slightly, minimizing cumulative energy loss to aerodynamic drag.
- The flatter trajectory keeps thrust vectoring tightly aligned with horizontal velocity.

Key Engineering Insight

Accounting for atmospheric drag forces ($F_{drag} \propto v^2$), reducing the launch angle from 45° to ~35° maximizes horizontal range by converting more internal pneumatic energy directly into forward momentum while mitigating drag accumulation.

4. Custom PVC Launch Stand & System Setup

The launch mechanism utilizes a rigid PVC pipe structure tailored specifically for stable field operation:

- **Structure & Rigidity:** Built using durable PVC pipes and joints forming a triangular/A-frame support bed to absorb recoil forces during release.
- **Angle Alignment:** Pre-fixed geometry ensures consistent launch angle (~35°) prior to pressurization.
- **Release & Safety:** Integrated pressure-locking quick-release nozzle mechanism operated via a safety tether, ensuring complete operator safety during high-pressure launches.

5. Field Test Results & Conclusion

During field testing on open grass grounds, the rocket demonstrated clean aerodynamic separation from the PVC launch stand, immediate water expulsion, and a stable parabolic flight path with minimal pitching or yaw instability.

- **Flight Profile:** Smooth ascent phase under water-powered thrust followed by a long, flat ballistic glide path.
- **Landing Outcome:** Successful recovery in the field downrange without structural deformation to the nose cone or fin assembly.
- **Conclusion:** The combination of a 35° launch angle, 25% propellant volume fraction, and custom PVC stand verified the aerodynamic model, achieving optimal ground distance.